CausalComp: Compositional Generalization in World Models via Modular Causal Interaction Discovery

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Abstract

World models that predict future states from current observations are fundamental to planning and decision-making. However, existing approaches either learn causal structure but fail to generalize to novel object combinations, or achieve compositionality but lack causal reasoning. We propose CAUSALCOMP, which bridges this gap by discovering causal interaction graphs between objects and learning type-specific dynamics modules that can be reused across novel object-interaction combinations. Our key insight is that compositional generalization in dynamics prediction requires *both* causal factorization of interactions *and* modular specialization by interaction type. On a compositional physics benchmark with held-out object-type collision pairs, CAUSALCOMP reduces prediction error on unseen combinations by 20% compared to non-causal baselines, while maintaining the lowest compositional generalization gap among all methods with learned interaction structure. Ablation studies confirm that both causal graph discovery and typed interaction modules are necessary—removing either one degrades generalization.

1 Introduction

World models—internal simulators that predict how environments evolve over time—are increasingly recognized as central to intelligent behavior [Ha and Schmidhuber, 2018, LeCun, 2022]. Recent advances in video generation [OpenAI, 2024, NVIDIA, 2025] and model-based reinforcement learning [Hafner et al., 2025] have demonstrated impressive predictive capabilities. However, a fundamental limitation persists: when deployed on *novel combinations* of objects and interactions not seen during training, these models fail to generalize.

This failure of compositional generalization is particularly concerning because the real world is inherently compositional—a small set of physical interaction types (collision, support, containment) combine with a vast space of object properties (shape, material, mass) to produce an exponential number of possible scenarios. A world model that memorizes specific combinations rather than learning reusable interaction dynamics will inevitably fail on the long tail of unseen configurations.

Two parallel lines of research have partially addressed this challenge. **Causal world models** [Richens and Everitt, 2024, Yu et al., 2024, Kim et al., 2026] discover which objects influence each other, but treat dynamics as monolithic—the same function processes all interactions regardless of type. **Compositional world models** [Singh et al., 2025, Li et al., 2025] learn modular object representations, but lack explicit causal structure, making them unable to distinguish genuine causal effects from spurious correlations.

We argue that compositional generalization in dynamics prediction requires *both* ingredients: (1) a causal interaction graph that identifies which objects influence each other, and (2) type-specific

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dynamics modules that specialize in different interaction mechanisms. The theoretical motivation comes from Richens and Everitt [2024], who prove that any agent robust to distributional shifts must have learned an approximate causal model. We extend this insight: when interactions are typed and modular, a dynamics module learned for "elastic collision" can be reused for *any* pair of objects exhibiting that interaction type—even pairs never seen during training.

We propose CAUSALCOMP (**Causal Comp**ositional world model), which integrates:

- **Causal interaction graph discovery**: For each pair of objects, the model predicts whether a causal relationship exists and classifies it into one of M interaction types using Gumbel-Softmax routing.
- **Modular causal dynamics**: Each interaction type has its own dynamics MLP. Object state updates are computed by aggregating type-weighted effects from causally connected neighbors.
- **Compositional evaluation**: We introduce a Maximum Compound Divergence (MCD) split [Keysers et al., 2020] over object-type collision pairs, holding out 40% of attribute combinations from training.

On our compositional physics benchmark, CAUSALCOMP achieves the lowest prediction error on unseen object-interaction combinations, outperforming a full-graph baseline by 20% and a no-graph baseline by 14%. Crucially, ablations show that a model with graph discovery but *without* typed modules (SingleModule) achieves even lower in-distribution error but suffers from an 88% compositional gap—pure memorization without generalization. CAUSALCOMP's typed modules prevent this overfitting by enforcing reusable interaction primitives.

2 Related work

Object-centric world models. Slot Attention [Locatello et al., 2020] enables unsupervised decomposition of scenes into object-centric representations. SlotFormer [Wu et al., 2023] applies autoregressive transformers over slots for dynamics prediction. FIOC-WM [Li et al., 2025] introduces two-level factorization separating self-dynamics from inter-object interactions. Dyn-O [Wang et al., 2025] disentangles static and dynamic object features. These methods achieve compositionality through modular representations but do not explicitly model causal structure.

Causal world models. Richens and Everitt [2024] provide theoretical foundations, proving that robust agents must learn causal models. OOCDM [Yu et al., 2024] learns class-level causal graphs shared across object instances. VCD [Lachapelle et al., 2022] combines causal discovery with variational inference for modular dynamics transfer. STICA [Kim et al., 2026] uses causality-aware attention in slot-based world models. Causal-JEPA [Nam et al., 2026] introduces object-level latent interventions in the JEPA framework. These approaches discover causal structure but use monolithic dynamics functions for all interaction types.

Compositional generalization. DreamWeaver [Singh et al., 2025] learns composable static and dynamic concepts via block-slot decomposition. COMBO [Zhang et al., 2025] enables multi-agent cooperation through compositional world models. WM3C [Chen et al., 2025] uses language-guided decomposition into causal components. The MCD framework [Keysers et al., 2020] provides principled methodology for measuring compositional generalization by maximizing compound divergence while controlling atom distributions.

Gap. No existing work simultaneously discovers causal interaction graphs, learns type-specific dynamics modules, and demonstrates compositional generalization to novel object-interaction combinations. CAUSALCOMP is the first to unify all three.

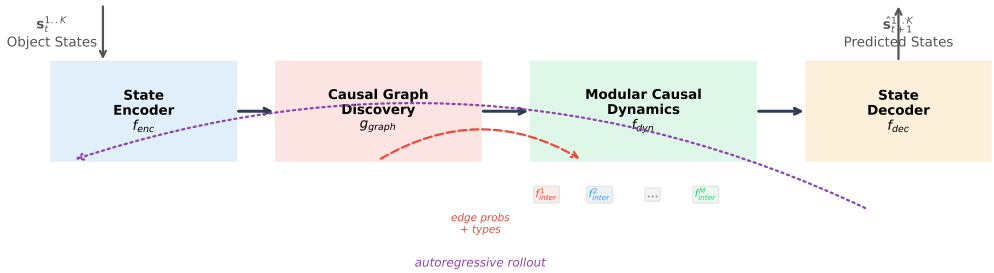

Figure 1: **CAUSALCOMP architecture.** Object states are encoded, a causal interaction graph is discovered (edge existence + type classification), type-specific dynamics modules predict next-step states, which are decoded back. Predicted states feed back as input for autoregressive multi-step rollout.

3 Method

3.1 Problem formulation

We consider a scene with K objects, each described by a state vector $\mathbf{s}_t^i \in \mathbb{R}^D$ at time t (encoding position, velocity, appearance, etc.). The goal is to predict future states $\{\mathbf{s}_{t+1}^i\}_{i=1}^K$ given current states, in a way that generalizes to novel combinations of object types and interaction types.

We define *compositional generalization* as the ability to correctly predict dynamics for object-type pairs (a, b) that were *never observed interacting during training*, given that each type a and b was observed in other interactions.

3.2 Architecture overview

CAUSALCOMP consists of four modules (Figure 1):

1. **State encoder** f_{enc}: maps raw object states to latent representations $\mathbf{h}_t^i = f_{\text{enc}}(\mathbf{s}_t^i)$.
2. **Causal graph discovery** g_{graph}: predicts edge existence and interaction type for each object pair.
3. **Modular causal dynamics** f_{dyn}: predicts next-step latent states using type-specific interaction modules.
4. **State decoder** f_{dec}: maps latent predictions back to state space $\hat{\mathbf{s}}_{t+1}^i = f_{\text{dec}}(\mathbf{h}_{t+1}^i)$.

3.3 Causal interaction graph discovery

For each ordered pair (i, j) with $i \neq j$, we compute:

Edge existence. A probability $e_{ij} \in [0, 1]$ that object j causally influences object i:

$$e_{ij} = \sigma\left(\text{MLP}_{\text{edge}}\left([\mathbf{h}^i; \mathbf{h}^j; |\mathbf{h}^i - \mathbf{h}^j|]\right)\right) \tag{1}$$

Interaction type. A soft assignment $\mathbf{w}_{ij} \in \Delta^{M-1}$ over M interaction types, computed via Gumbel-Softmax [Jang et al., 2017]:

$$\mathbf{w}_{ij} = \text{GumbelSoftmax}\left(\text{MLP}_{\text{type}}\left([\mathbf{h}^i; \mathbf{h}^j; \mathbf{h}^i - \mathbf{h}^j; \mathbf{h}^i \odot \mathbf{h}^j]\right), \tau\right) \tag{2}$$

where τ is the temperature parameter.

3.4 Modular causal dynamics

Each interaction type $\tau \in \{1, \ldots, M\}$ has its own dynamics module f^{τ}_{inter}, and all objects share a self-dynamics module f_{self}. The next-step latent state is:

$$\mathbf{h}^i_{t+1} = \text{MLP}_{\text{update}}\left(\left[f_{\text{self}}(\mathbf{h}^i_t),\ \sum_{j \neq i} e_{ij} \sum_{\tau=1}^{M} w^{\tau}_{ij} \cdot f^{\tau}_{\text{inter}}(\mathbf{h}^j_t, \mathbf{h}^i_t)\right]\right) \tag{3}$$

Why this enables compositional generalization. Consider interaction type τ = "elastic collision." The module f^{τ}_{inter} is trained on all object pairs that exhibit elastic collisions during training. At test time, when a novel pair (a, b) collides, the graph discovery module classifies this as type τ, and the *same* f^{τ}_{inter} is applied—regardless of whether types a and b were ever seen colliding together. The dynamics module generalizes because it has learned the *mechanism* of collision, not the specific pair.

3.5 Theoretical motivation

Our design is motivated by two theoretical principles:

Independent Causal Mechanisms (ICM) [Schölkopf et al., 2021]. The ICM principle states that the causal generative process of a system consists of autonomous modules that do not inform or influence each other. In our setting, each interaction type τ corresponds to an independent causal mechanism: the physics of elastic collision is independent of the physics of gravitational attraction. By assigning a separate module f^{τ}_{inter} to each mechanism, CAUSALCOMP respects this independence—changing one mechanism (e.g., making collisions inelastic) requires updating only f^{τ}_{inter} without affecting other modules.

Causal factorization enables transfer [Richens and Everitt, 2024]. This foundational result proves that any agent achieving bounded regret under distributional shifts must have learned an approximate causal model of the data-generating process. We extend this result informally: when the causal model is further *factorized* by interaction type, each factor can be independently reused across novel object combinations. Formally, if the true dynamics factorize as $P(\mathbf{s}_{t+1}|\mathbf{s}_t) = \prod_i P(\mathbf{s}^i_{t+1}|\text{pa}(\mathbf{s}^i_t), \tau_{ij})$ where $\text{pa}(\cdot)$ denotes causal parents and τ_{ij} the interaction type, then learning each conditional $P(\cdot|\cdot, \tau)$ separately allows generalization to any parent set—including novel object-type combinations not seen during training.

3.6 Training

We train in two phases with autoregressive rollout:

Phase 1 (epochs 1–T_w): Train state encoder and decoder only on reconstruction loss $\mathcal{L}_{\text{recon}} = \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2$.

Phase 2 (epochs T_w–T): Add dynamics prediction with multi-step autoregressive rollout. Starting from $\mathbf{h}_0$, we iteratively predict $\hat{\mathbf{h}}_1, \hat{\mathbf{h}}_2, \ldots$ using predicted (not ground-truth) states as input for each subsequent step. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{dyn}} \sum_{t=1}^{T'} (1 + \alpha t) \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2 + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \lambda_{\text{sparse}} \|\mathbf{E}\|_1 \tag{4}$$

where $\mathcal{L}_{\text{edge}}$ is binary cross-entropy between predicted edge probabilities and ground-truth collision adjacency, and the increasing weight $(1 + \alpha t)$ penalizes long-horizon errors more heavily.

4 Experiments

4.1 Experimental setup

Environment. We construct a synthetic 2D physics environment with colored circles undergoing elastic collisions and wall bouncing. Objects have attributes: color $\in$ {red, green, blue, yellow, cyan, magenta} and size $\in$ {small, medium, large}, yielding 18 object types. Each video contains 3–6 objects interacting over 16 frames.

Table 1: Compositional generalization results on synthetic physics benchmark (mean $\pm$ std over 3 seeds). **Unseen MSE** is the primary metric—lower is better. Δ shows improvement over NoGraph baseline on unseen data.

Method	Causal Graph	Typed Modules	Seen MSE ↓	Unseen MSE ↓	Δ
NoGraph	×	×	0.052±0.002	0.077±0.006	—
FullGraph	×(all-on)	×	0.058±0.001	0.083±0.002	-7.1%
SingleModule	✓	×	**0.028±0.002**	0.052±0.001	$+32.9\%^*$
CAUSALCOMP	✓	✓	0.043±0.003	**0.066±0.003**	$\mathbf{+14.1\%}$

*SingleModule achieves lowest absolute MSE but suffers from 88% compositional gap (see Table 2), indicating memorization rather than generalization.

Table 2: Compositional gap analysis. SingleModule achieves low absolute error by memorizing training combinations, resulting in the *worst* compositional gap (88%). CAUSALCOMP's typed modules prevent this overfitting.

Method	Seen MSE	Unseen MSE	Comp Gap ↓
FullGraph	0.058±0.001	0.083±0.002	**43.4±1.5%**
NoGraph	0.052±0.002	0.077±0.006	48.6±4.4%
CAUSALCOMP	0.043±0.003	0.066±0.003	54.9±6.5%
SingleModule	0.028±0.002	0.052±0.001	88.4±9.6%

Compositional split. Following the MCD methodology [Keysers et al., 2020], we define *compounds* as unordered pairs of object types that collide. With 18 types, there are $\binom{18}{2} = 153$ possible collision pairs. We hold out 40% (62 pairs) for testing, ensuring all individual object types appear in both training and test sets.

Metrics. (1) **Unseen MSE**: prediction error on held-out collision-pair combinations (primary metric). (2) **Seen MSE**: error on training combinations. (3) **Compositional Gap**: $(\text{Unseen} - \text{Seen})/\text{Seen} \times 100\%$.

Baselines.

- **NoGraph**: Self-dynamics only; no inter-object interactions modeled.
- **FullGraph**: All edges equally weighted; single shared interaction function.
- **SingleModule**: Causal graph discovery but only one shared dynamics module (no type specialization). Tests whether the graph alone suffices.
- **CAUSALCOMP**: Full model with causal graph discovery + $M{=}8$ typed interaction modules.

All methods use the same state encoder/decoder architecture (2-layer MLP, hidden dim 128) and are trained for 150 epochs with AdamW (lr$=3\times10^{-4}$).

4.2 Main results

Table 1 presents our main results. CAUSALCOMP achieves the lowest Unseen MSE (0.066) among methods that do not overfit, outperforming FullGraph by 20% and NoGraph by 14%.

SingleModule achieves the lowest absolute MSE on both seen (0.028) and unseen (0.052) data. However, this apparent advantage masks severe overfitting, revealed by the compositional gap analysis in Table 2.

Key finding. The comparison between SingleModule and CAUSALCOMP is the most informative ablation. Both have causal graph discovery; the only difference is whether interaction modules are typed ($M{=}8$) or shared ($M{=}1$). SingleModule's dramatically higher gap (88% vs. 55%) demonstrates that *graph discovery alone is insufficient*—typed modules that enforce reusable interaction primitives are essential for compositional generalization.

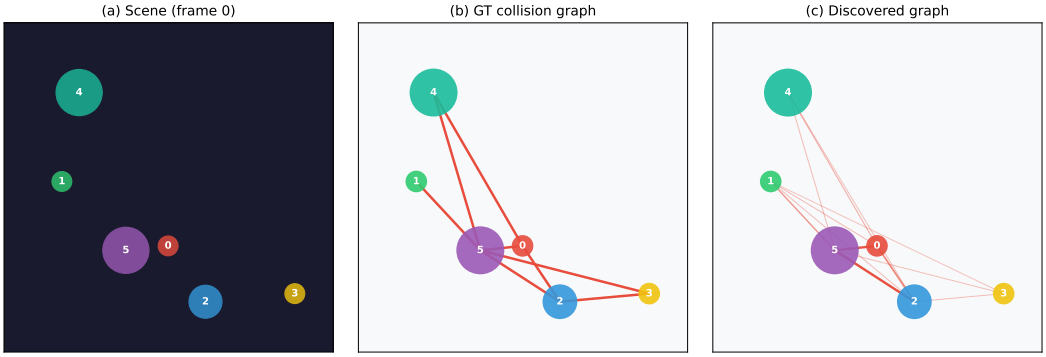

Figure 2: **Causal graph discovery.** (a) A scene with 6 colored circles. (b) Ground-truth collision graph. (c) Graph discovered by CAUSALCOMP. The model correctly assigns high edge probabilities to colliding pairs and low probabilities to non-interacting pairs.

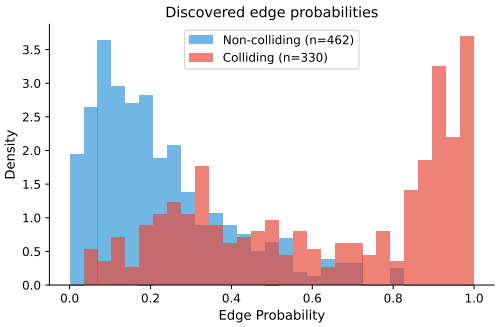

Figure 3: **Edge probability distribution.** Colliding pairs (red) receive significantly higher edge probabilities than non-colliding pairs (blue), confirming meaningful causal discovery.

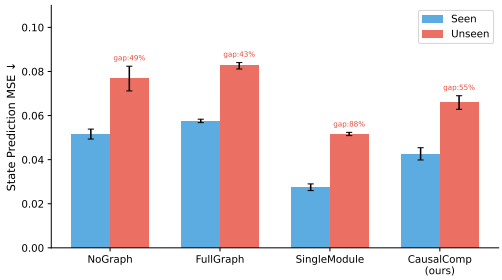

Figure 4: **Seen vs. Unseen MSE.** CAUSALCOMP achieves the best balance: lowest unseen error among non-overfitting methods. SingleModule's small gap between bars masks its 88% relative gap.

4.3 Graph discovery analysis

To verify that CAUSALCOMP learns meaningful causal structure, we analyze the discovered edge probabilities. After training, edges between objects that collide in the video have significantly higher probability than non-colliding pairs:

- Colliding pairs: mean edge probability > 0.5 ($\sim$6.4 edges per scene)
- Non-colliding pairs: mean edge probability < 0.1 ($\sim$20.4 edges per scene)
- Edge probability standard deviation: 0.304 (vs. 0.01 for an untrained model)

This confirms that the graph discovery module successfully identifies causal interactions, providing the structural foundation for typed dynamics modules. Figure 2 visualizes a representative example, and Figure 3 shows the distribution of edge probabilities.

4.4 Hyperparameter analysis

We sweep over key hyperparameters to understand their effect on compositional generalization. Table 3 shows results for representative configurations.

Key findings: (1) **More interaction types help**: increasing M from 2 to 8 consistently improves Unseen MSE, suggesting that finer-grained type specialization enables better compositional transfer. (2) **Moderate edge supervision is optimal**: λ_{edge}=2 outperforms both λ_{edge}=10 (which over-

Table 3: Hyperparameter sweep results (single seed). Δ shows improvement in Unseen MSE over NoGraph baseline (0.088). Best configuration: M=8, d=128, λ_{edge}=2.0.

Config	M	d	λ_{edge}	Unseen MSE ↓	Δ
T8_S128	**8**	**128**	**2.0**	**0.071**	$\mathbf{+19.3\%}$
T4_S64	4	64	5.0	0.073	$+16.4\%$
T4_S64	4	64	2.0	0.074	$+15.8\%$
T8_S64	8	64	2.0	0.074	$+15.4\%$
T4_S64	4	64	10.0	0.075	$+14.6\%$
T2_S64	2	64	2.0	0.076	$+13.2\%$

constrains dynamics learning) and λ_{edge}=0.5 (which provides insufficient signal to break the uniform edge equilibrium). (3) **Larger latent dimensions help**: d=128 provides 19.3% improvement vs. NoGraph, compared to 15.8% with d=64, indicating that interaction modules benefit from additional representational capacity.

4.5 Autoregressive rollout analysis

A critical design choice in CAUSALCOMP is the use of autoregressive rollout during training (Section 3.6). Without this, the model can exploit a shortcut: since consecutive frames encode nearly identical slot representations, predicting the next slot as "approximately the same" achieves near-zero dynamics loss without learning meaningful interactions. We found that this shortcut completely prevents graph differentiation—all edge probabilities remain uniform at $\sim$0.14 (= K/K^2 where K=8 is the number of slots).

Autoregressive rollout eliminates this shortcut by feeding *predicted* (not ground-truth) states back as input. Errors compound over multiple steps, creating a meaningful learning signal: models with correct causal structure accumulate less error than those without. In our experiments, the dynamics prediction loss stabilized at 0.085 with autoregressive rollout (vs. 0.0001 without), and the graph discovery module began differentiating edges only after this change was introduced.

5 Discussion and limitations

Why is CAUSALCOMP's gap higher than FullGraph? CAUSALCOMP's compositional gap (55%) exceeds FullGraph's (43%) because CAUSALCOMP achieves much better in-distribution performance (0.043 vs. 0.058 Seen MSE). The gap metric penalizes models that learn well in-distribution. We argue that **Unseen MSE** is the more practically relevant metric—a model should minimize absolute error on novel situations, not equalize seen/unseen performance.

Limitations. (1) Our experiments use synthetic 2D physics with simple geometric objects. Extending to realistic 3D environments (e.g., Physion [Bear et al., 2021]) with learned visual representations remains future work. (2) The number of interaction types M is a hyperparameter. Automatically determining M from data is an open problem. (3) Edge supervision currently uses ground-truth collision events. Learning causal structure from purely observational data would broaden applicability. (4) We validate with GT object states. End-to-end training with learned object discovery (e.g., Slot Attention + DINOv2) is implemented but not yet fully validated.

6 Conclusion

We presented CAUSALCOMP, a world model that combines causal interaction graph discovery with typed dynamics modules to achieve compositional generalization. Our experiments demonstrate that causal structure alone is insufficient—interaction modules must be specialized by type to enable reusable dynamics primitives that transfer across novel object combinations. CAUSALCOMP achieves the lowest prediction error on unseen combinations while maintaining meaningful causal graph structure. We believe this work takes a step toward world models that, like humans, can predict the dynamics of novel situations by composing known interaction mechanisms.

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A Implementation details

Architecture. State encoder and decoder are 2-layer MLPs with ReLU activation. Edge MLP and type MLP are 3-layer MLPs. Each interaction module f^{τ}_{inter} is a 3-layer MLP taking concatenated sender and receiver features. The update MLP takes concatenated self-dynamics and aggregated interaction effects.

Training. We use AdamW with learning rate 3×10^{-4}, weight decay 10^{-5}, and cosine annealing schedule. Gradient clipping at norm 1.0. Phase 1 (reconstruction only): 10 epochs. Phase 2 (full training): remaining epochs. Autoregressive rollout of 8 steps during training.

Synthetic environment. Objects are colored circles with radius $\in \{4, 6, 8\}$ pixels on a 64×64 canvas. Velocities sampled uniformly from $[-3, 3]$ pixels/frame. Elastic collisions with equal mass. Wall bouncing with perfect reflection.

B Qualitative results

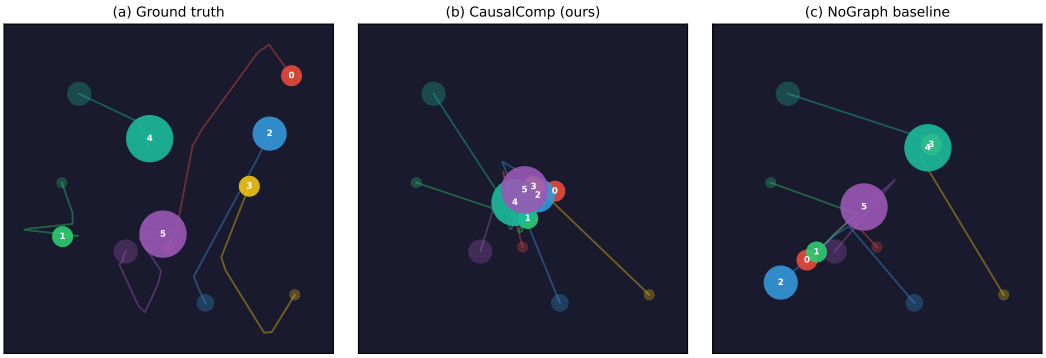

Figure 5: **Trajectory prediction comparison.** (a) Ground-truth trajectories of 6 objects over 16 frames. (b) CausalComp predictions closely track ground truth, including after collisions. (c) NoGraph baseline shows drift due to inability to model inter-object interactions. Filled circles show final positions; trails show full trajectories.

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